

Heterogeneous effects of borrower-based macroprudential policy

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Research questions

- ▶ How do borrower-based measures (BBMs) in the macroprudential toolkit affect the transition to home ownership and consumption/saving?
- ▶ How do they affect different age groups?
- ▶ To which extent do the quantitative effects depend on the incidence of renters in the housing market?
(as, e.g., determined by transaction costs or renting of housing services being a good substitute to owning)

Common perspective

- ▶ Tighter financing constraints prevent the young to become homeowners
- ▶ Macroprudential regulation has to trade off financial stability with welfare cost for the young, prospective home buyers

→ True for economies with a high incidence of renters?

Approach

- ▶ Dynamic portfolio choice model with housing market
 - ▶ Demand side
 - ▶ dynamic, stochastic, life-cycle model calibrated to Germany
 - ▶ continuous portfolio choice (housing asset, financial asset)
 - ▶ discrete home ownership decision featuring transaction costs
 - ▶ Supply side
 - ▶ schedule of housing units (rental or ownership)
 - ▶ units supplied by deep pocket investors, including new units through construction
 - ▶ small open economy: given interest rate
- ▶ Use model as a lab
 - ▶ policy experiments
 - ▶ explore heterogeneous effects, e.g., across ages, housing tenure status

Different perspective

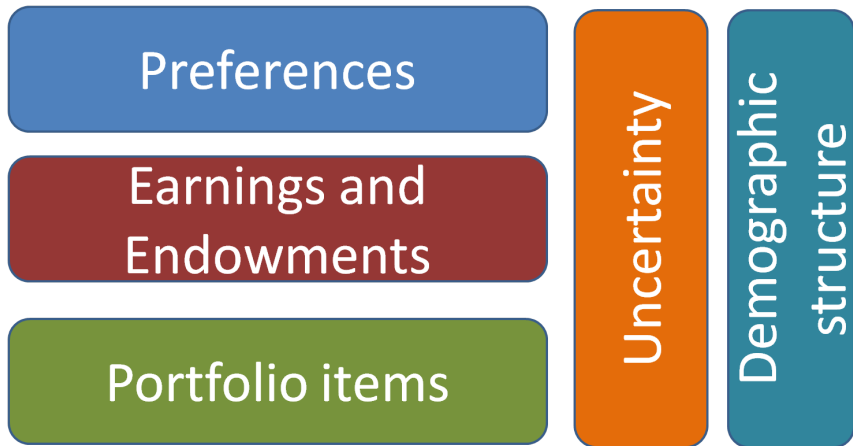
- ▶ Remaining a renter is less bad (in welfare terms) in a Germany-type economy than in a US-type economy
 - ▶ smaller cost for the young if they do not become homeowners
- ▶ Tighter financing constraints cool down the housing market
 - ▶ lower house prices hurt homeowners and benefit young renters, who are prospective home buyers

→ Macroprudential BBMs have to trade off financial stability with welfare cost for homeowners (who tend to be older)

Related Literature

- ▶ **Effect of borrower-based macroprudential policy:** Aastveit et al. (2024), Acharya et al. (2022), Balke et al. (2024), Bolliger et al. (2024), Castellanos et al. (2024), Cerasi et al. (2026), Fuster and Zafar (2016), Higgins (2025)
- ▶ **Heterogeneous consumption response to house-prices changes:** Berger et al. (2018), Kaplan, Mitman, Violante (2019), Mian and Sufi (2011) and Mian, Rao and Sufi (2013)
- ▶ **Heterogeneity in monetary-policy transmission and the housing market:** Calza, Monacelli, Stracca (2013), Corsetti, Duarte, Mann (2018), Hintermaier and Koeniger (2024)
- ▶ **Housing markets in Germany:** Kaas, Korchakov, Preugschat, Siassi (2021)

Model overview for demand side



Model blocks: demand side

Details
▽

Block 1: Preferences

Continuous choice:

Life-cycle discounted utility from non-durable consumption and housing services

Discrete choice:

Housing services from owned housing or rented housing

Block 2: Earnings and Endowments

Earnings profile

[Estimated from HFCS data]

Block U: Uncertainty

Idiosyncratic uncertainty of earnings

[Estimated from HFCS data]

Model blocks: demand side (ctd.)

Details
▽

Block 3: Portfolio items

Continuous portfolio choice:

- ▶ Liquid financial assets
allowing for interest-rate spread
between positive and negative holdings
- ▶ Illiquid housing asset: transaction costs

Liquidity

Debt is *collateralized* by housing &
restricted by (pdv of) labor earnings

Portfolio restrictions

Discrete choice:

Adjusting or not adjusting illiquid housing

Accounting for demographic composition

Block D: Demographic structure

Apply demographic composition

[Matching the composition of HFCS data]

Composing a synthetic survey,
using distributions from the model

Calibration (steady state)

Panel A: Externally calibrated parameters

μ_{adj}	0.8	loan-to-value ratio of adjuster before retirement
$\mu_{\text{no adj}}$	0.85	loan-to-value ratio of non-adjuster before retirement
$\mu_{\text{adj}}^{\text{ret}}$	0.2	loan-to-value ratio of adjuster after retirement
$\mu_{\text{no adj}}^{\text{ret}}$	0.25	loan-to-value ratio of non-adjuster after retirement
$\delta^{\dagger}$	0.15	loan-to-income factor
r^{+}	0.015	lending: real interest rate
r^{-}	0.03	borrowing: real interest rate
r	0.015	long-run real interest rate, applied to bequests
g	0.01	aggregate income growth rate
$\underline{k}$	0.0125	non-interest component of rent-to-price ratio
α_1	0.016	proportional transaction cost for selling housing
α_2	0.09	proportional transaction cost for buying housing
σ	0.01	scale parameter of taste shock for discrete choice
ρ	0.95	autocorrelation of income shocks
further inputs	life-cycle age profiles of income and income risk pensions, tax systems and minimum income benefits age distribution and survival probabilities beginning-of-life-cycle asset distribution	

Panel B: Internally calibrated preference parameters

	Preference types					
	type 1	type 2	type 3	type 4	type 5	
β	0.9	0.985	0.99	0.995	0.995	discount factor
θ	0.86	0.7	0.7	0.8	0.84	weight of non-housing consumption in utility function
ϕ_R	0.94	0.94	0.94	0.94	0.94	rental efficiency
weight	0.346	0.041	0.361	0.189	0.063	weight of type in the population

Model performance

	Data	Model
Net worth	201,304	200,313
of which: housing wealth	95,676	86,240
Renter share (%)	53.7	54.9
Mortgagor share (%)	9.4	12.1
Extended LTV (%)	3.3	3.3
Extended LTI (%)	21.0	23.1
Extended share LTV > 0.5 (%)	3.2	1.9
Extended share LTI > 1 (%)	6.5	7.0

Notes: Means in adult equivalents for households aged 26 – 75.
Mortgagor share denotes share with negative financial assets.

Supply side and equilibrium (preview)

- ▶ Supply side summarized by **quantity-price schedule**
- ▶ **Price elasticity** disciplined by reduced-form evidence on price responses to changes in transaction tax
- ▶ **Market clearing** along transition to new steady state
- ▶ Solved numerically using **shooting method** based on price sequence (Boppart et al., 2018)
- ▶ Approach allows for **market segmentation**

Responses of home ownership and consumption

Quantify the responses to the following changes:

1. Financing constraints:

max LTV tighter by 20 pp,

max LTI tighter by 1/3

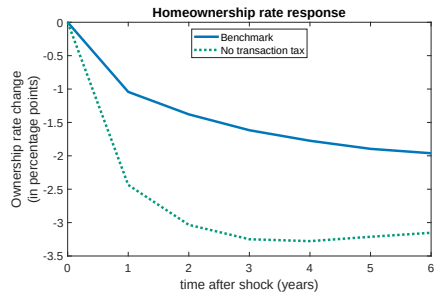
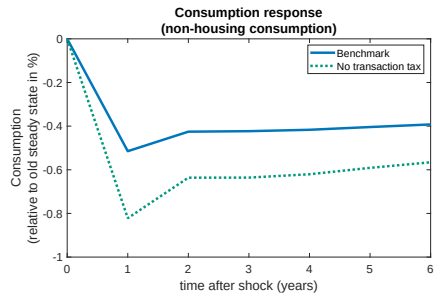
2. House price drop:

House price decreases by 10 percent

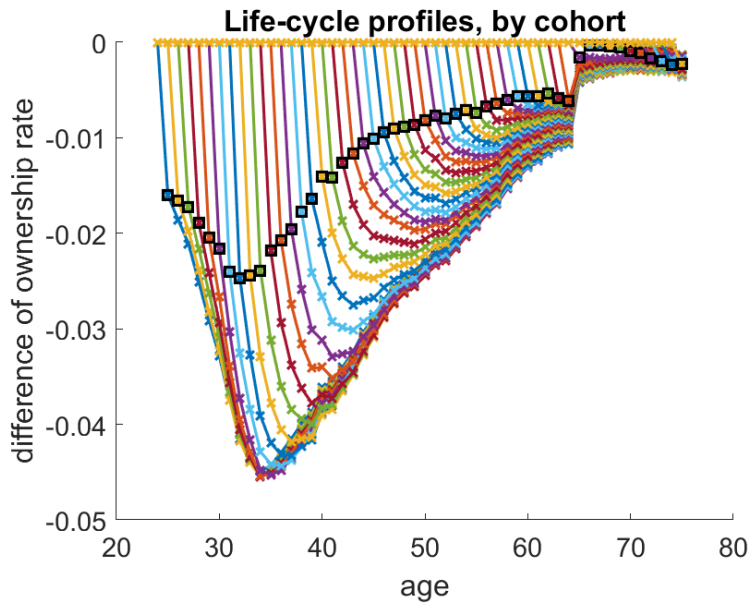
Impact responses

Experiments						
<i>Decrease of ...</i>						
Response	max. LTV, house price	max. LTI, house price	max. LTV, LTI, house price	max. LTV	max. LTI	house price
<i>Benchmark</i>						
Consumption (in percent)	-0.55	-0.45	-0.51	-0.03	0.05	-0.50
Homeownership (in perc. points)	-0.77	-0.56	-1.04	-0.47	-0.45	-0.10
<i>No transaction tax</i>						
Consumption (in percent)	-0.83	-0.69	-0.82	-0.10	0.04	-0.73
Homeownership (in perc. points)	-2.38	-0.70	-2.44	-1.45	-0.44	-0.26

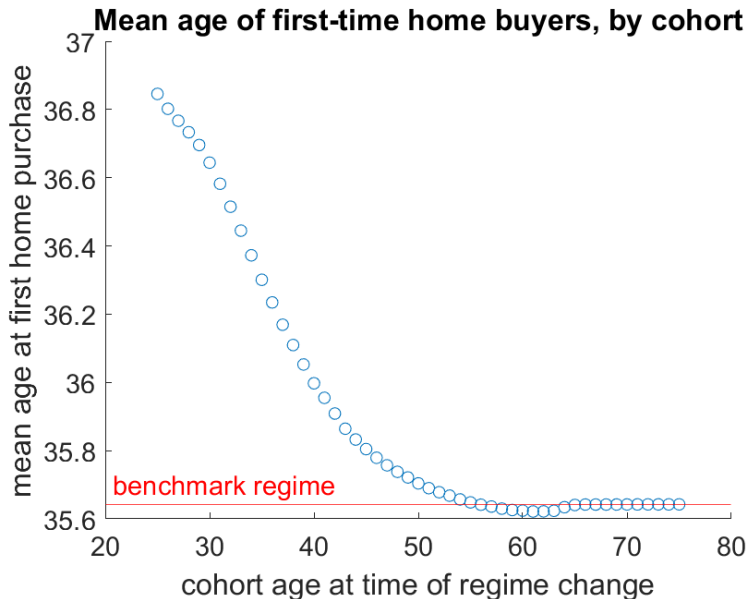
Responses with and without transaction tax



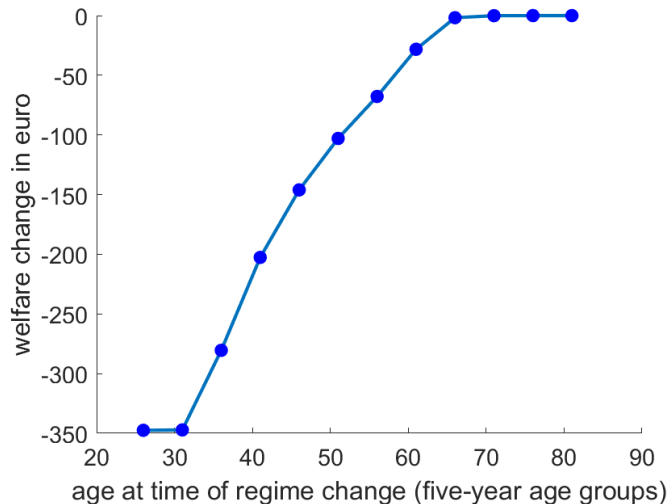
Cohort effects of tighter financing constraints



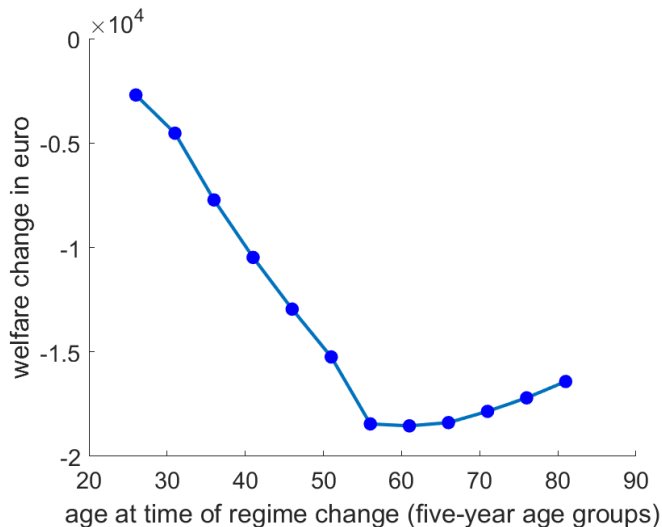
Effects on average age of 1st-time home buyers



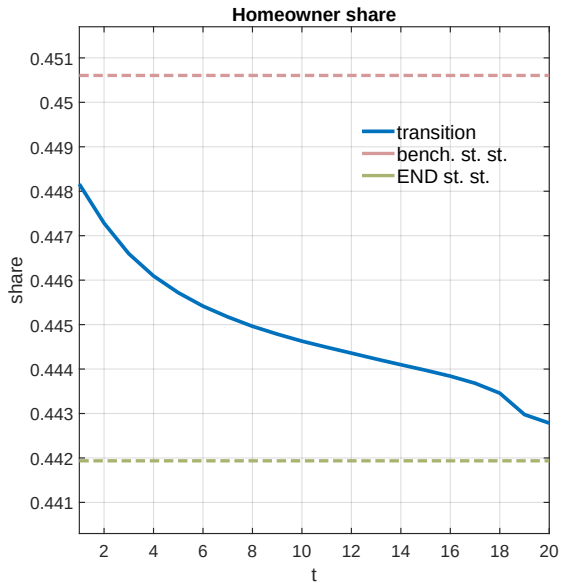
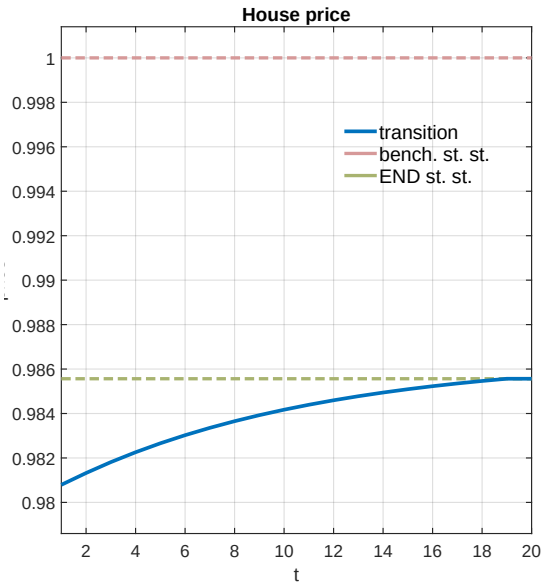
Welfare effect of tighter financing constraints by age cohort *holding house prices constant*



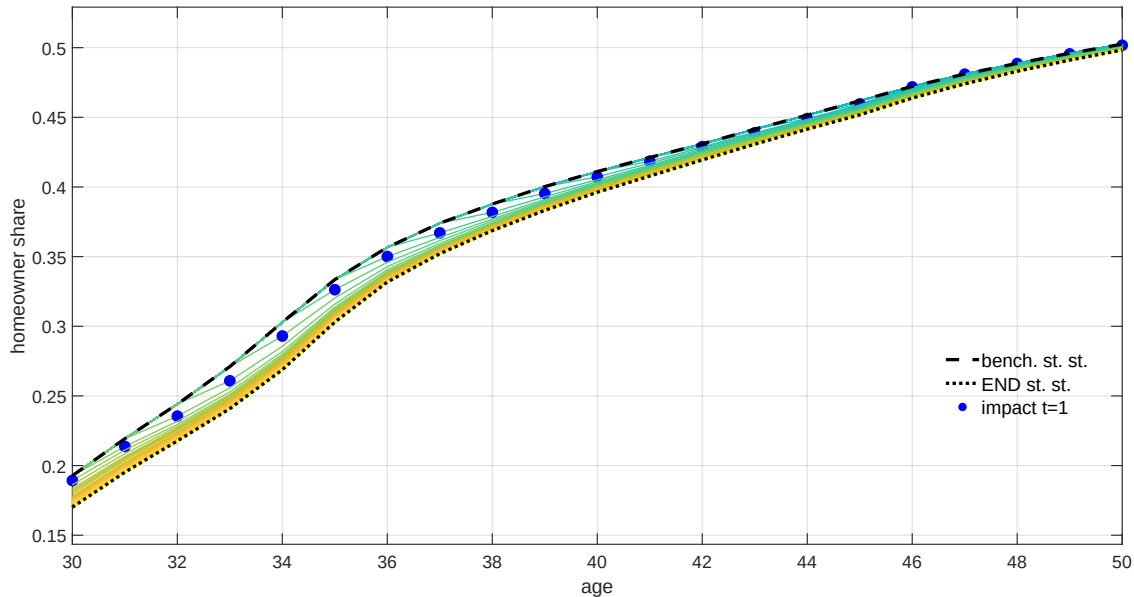
Welfare effect of tighter financing constraints, by age cohort *with associated house-price drop*



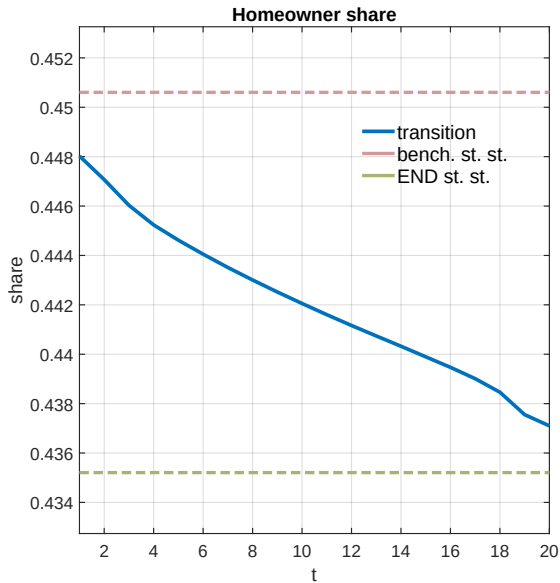
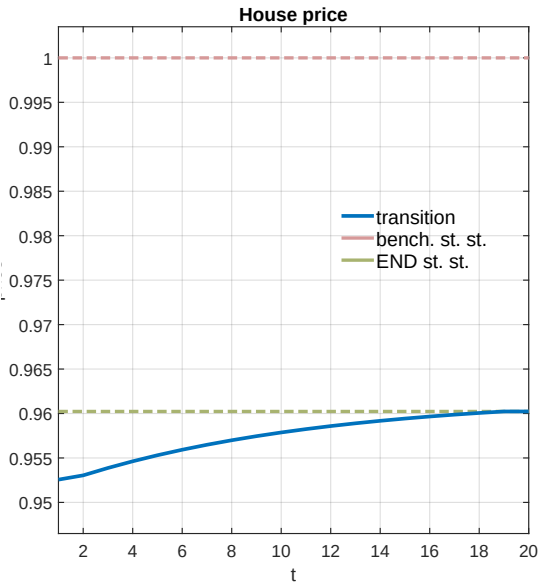
Endogenous price response: low LTV



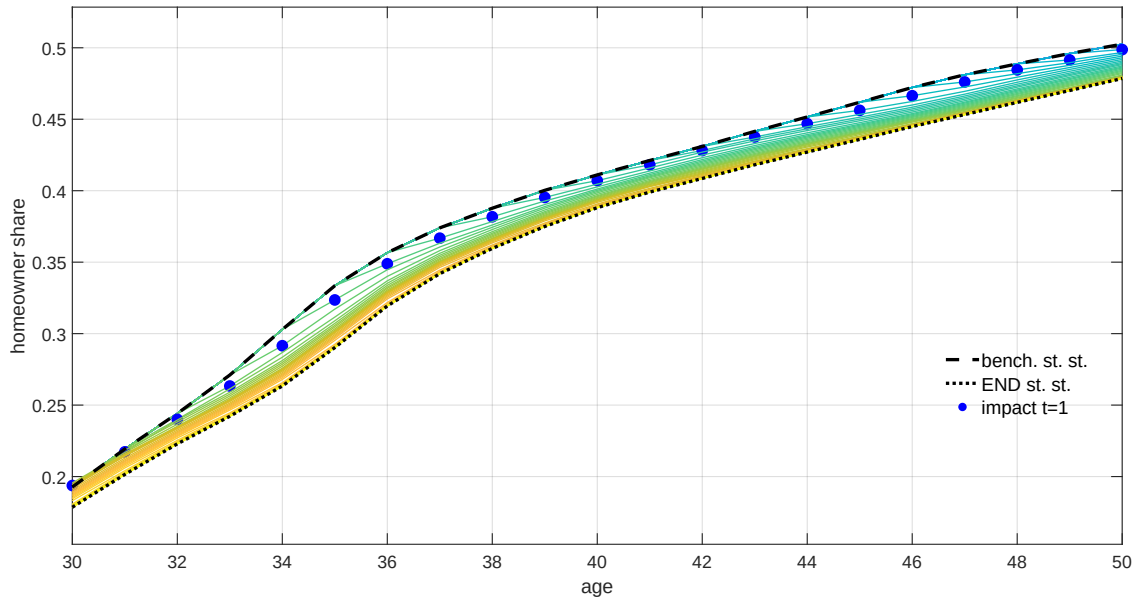
Endogenous price response: low LTV, homeowner share profiles by cohort



Endogenous price response: low LTI



Endogenous price response: low LTI, homeowner share profiles by cohort



Conclusions

- ▶ Trade-offs implied by borrower-based macroprudential measures depend on features of the housing market such as transaction costs or rental efficiency
- ▶ Welfare effect of a regime with tighter financing constraint is dominated by the associated reduction of house prices (for empirically plausible orders of magnitude)
- ▶ Home owners, who tend to be older, bear most of the welfare costs
- ▶ Younger agents, who are more likely to be renters, delay their transition to home ownership
- ▶ This is not too costly in welfare terms if rented housing is a good substitute for owned housing

Model details



Life-cycle problem of the household

$$\mathbb{E}_0 \left[\sum_{j=1}^J \beta^{j-1} \left((1 - \iota_j) u(c_j, \hat{s}_j) + \iota_j \Psi(a_j, h_j) \right) \right],$$

If choosing to consume housing as an **owner, adjusting his housing stock**:

$$\hat{f}_j = 0,$$

$$\hat{s}_j = \phi \hat{h}_{j+1},$$

the budget constraint

$$c_j + a_{j+1} + p_t \hat{h}_{j+1} + \alpha_{p,j}(\hat{h}_j, \hat{h}_{j+1}) = y_j + (1 + r_{t-1})a_j + p_t \hat{h}_j,$$

and the financing constraints

$$(1 + r_j) a_{j+1} \geq -\mu_{\Delta} p_t \hat{h}_{j+1},$$

$$(1 + r_j) a_{j+1} \geq -\tilde{g}_{y,j}.$$

Model details continued, adjustment cost $\triangle$

The adjustment cost function is

$$\alpha_{p,j}(\hat{h}_j, \hat{h}_{j+1}) = \alpha_1 p_t \hat{h}_j + \alpha_2 p_t \hat{h}_{j+1}$$

Model details continued, non-adjuster $\triangle$

If the household chooses to consume housing as an **owner, not adjusting** his housing stock:

$$\hat{h}_{j+1} = \hat{h}_j,$$

$$\hat{f}_j = 0,$$

$$\hat{s}_j = \phi \hat{h}_j,$$

the budget constraint

$$c_j + a_{j+1} = y_j + (1 + r_{t-1})a_j,$$

and the financing constraints

$$(1 + r_j) a_{j+1} \geq -\mu_{\Delta} p_t \hat{h}_j,$$

Model details continued, renter $\triangle$

*If the household chooses to consume housing as a **renter**:*

$$\hat{h}_{j+1} = 0,$$

$$\hat{s}_j = \phi_R \hat{f}_j,$$

the budget constraint

$$c_j + a_{j+1} + q_t \hat{f}_j + \alpha_{pR}(\hat{h}_j) = y_j + (1 + r_{t-1})a_j + p_t \hat{h}_j,$$

and the financing constraint

$$(1 + r_t)a_{j+1} \geq 0.$$

Model details: recursive formulation $\triangle$

Define liquidable wealth:

$$x_j = (1 + r_{t-1}) a_j + (1 - \alpha_1) h_j$$

Bellman equation:

$$W_j(x_j, h_j, s_j) = \max_{d_j, c_j, f_j, a_{j+1}, h_{j+1}} \left[U(c_j, \bar{s}_j) + (1 - \iota_j) \beta \sum_{s_{j+1} \in S} \pi_{s_j, s_{j+1}} W_{j+1}(x_{j+1}, h_{j+1}, s_{j+1}) + \iota_j \Psi(x_{j+1}) \right],$$

subject to constraints

- ▶ discrete choice at age j : $d_j \in \{\text{owning-and-not-adjusting, owning-and-adjusting, renting}\}$
- ▶ discrete states $s \in S$ with transition probabilities $\pi_{s, s'}$
- ▶ $\bar{s}$: price-transformed service flow from housing
- ▶ ι_j : probability of death at the end of period j

LTV and LTI constraint $\triangle$

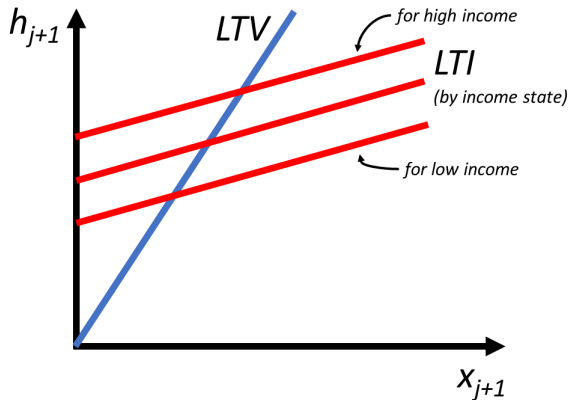


Figure: LTV constraint (blue) and LTI constraint (red)

The LTV and LTI constraints are depicted in the space of x (liquidable wealth) and h (housing), i.e. the variables used for the recursive solution. A family of LTI constraints is shown, illustrating the income-state dependence of the LTI constraint.

Age profiles

